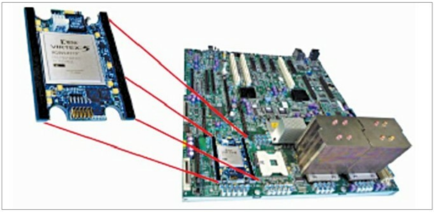

Fig. 3: 'In socket' coherent accelerators

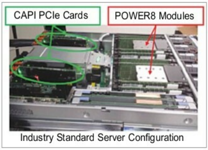

Fig. 4: Power8 modules with CAPI PCIe cards

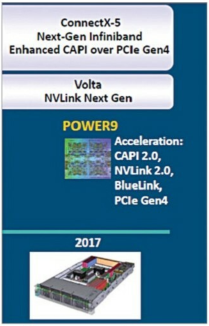

Fig. 5: Power processor roadmap

floating-point coprocessors, GPUs to accelerate the rendering of a vertex-based 3D model into a 2D viewing plane, and accelerators for the motion estimation step of a video codec. Applications such as big data analytics, search, machine learning, network functions virtualisation, wireless 4G/5G, in-memory database processing, video analytics and network processing benefit from acceleration engines, which need to move data seamlessly among the various system components.

Coherent accelerators

HyperTransport, front-side bus (FSB) and QuickPath Interconnect (QPI) are not commercially successful because these have a proprietary 'in socket' attached, server mechanical fit challenges, need to give up processor socket, proprietary specification, licensing issues and always-lagging 6-12 month Cadence.

IBM's Coherent Accelerator Processor Interface (CAPI) interface resolves all these issues. This OpenPower-based solution is well documented and already commercialised. It has proprietary 'in socket' attached, and processors use all the available sockets. It is based on PCIe, hence there are no mechanical challenges. It is free, simple, OpenPower-licensed. Also, it is in sync with Power and Cadence' PCIe interface IP.

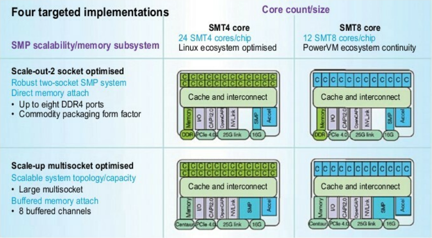

Fig. 6: SMT4 vs SMT8 core

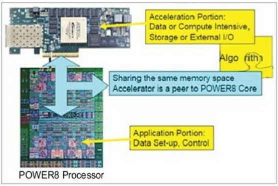

Fig. 7: CAPI FPGA accelerator (based on a standard PCIe accelerator)

Power9 processor

IBM's next-generation Power9 processor will have 24 cores, double that of today's Power8 chips. Design is optimised for two-socket scale-out servers (hence the name Power9 SO), and includes on-chip acceleration for compression and encryption.

Power9 will be available in two